1 Download of operation data

Structure E2BP_PP_PDC_OPERA2

Field name	T	L	D	Description	Usage in HYDRA
SOURCE_SYS	CHAR	10	0	Logical system	Not used
CONF_NO	NUMC	10	0	Confirmation/upload number of the operation	Confirmation/upload number
ORDERID	CHAR	12	0	Order	According to configuration (*1)
SEQUENCE	CHAR	6	0	Sequence	According to configuration (*1)
OPERATION	CHAR	4	0	Operation	According to configuration (*1)
SUB_OPER	CHAR	4	0	Suboperation	According to configuration (*1)
SUBSYSTEM_GROUPING	CHAR	3	0	Grouping of subsystem connection	Not used
MATERIAL	CHAR	18	0	Material	Order header, final item number, OP, item number
MATL_DESC	CHAR	40	0	Material description	Order header, final item description, OP item description
ROUT_QUAN_UNIT	UNIT	3	0	Quantity unit of the plan	Not used
ROUT_QUAN_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used
ORD_QUAN_UNIT	UNIT	3	0	Quantity unit for in-house production	Not used
ORD_QUAN_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used
BASE_QUAN_UNIT	UNIT	3	0	Base quantity unit	Base quantity unit; Entered in the order header for the first operation of an order. The operations "inherit" this base quantity unit from the order header. Please note: This unit must be the same for all operations of an order!
BASE_QUAN_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used

Field name	T	L	D	Description	Usage in HYDRA
HDR_NOMINATOR	DEC	5	0	Numerator for the conversion to base quantity unit Please note: conversion factors from ORD_QUAN_UNIT into BASE_QUAN_UNIT. If OPER_QUAN_UNIT <> ORD_QUAN_UNIT wrong calculations might be the result!!!	Numerator for the conversion from primary to HYDRA base quantity unit
HDR_DENOMINATOR	DEC	5	0	Denominator for the conversion to the base quantity unit	Denominator for the conversion from primary to HYDRA base quantity unit
LEADING_ORDERID	CHAR	12	0	Order network: Leading order	Not used
SUPERIOR_ORDERID	CHAR	12	0	Order network: Directly superior order	Not used
SUPERIOR_SEQUENCE	CHAR	6	0	Order no.: Sequence of the directly superior order	Not used
SUPERIOR_OPERATION	CHAR	4	0	Order network: Operation of the directly superior order	Not used
REFERENCE_SEQUENCE	CHAR	6	0	Parallel sequence: Reference sequence of a sequence	Reference sequence
BRANCH_OPERATION	CHAR	4	0	Parallel sequence: Branch operation	Branch operation of the parallel sequence
RETURN_OPERATION	CHAR	4	0	Parallel sequence: Return operation	Return operation of the parallel sequence
OPER_DESCRIPTION	CHAR	40	0	Short text of operation	OP, operation designation
OPER_QUAN_UNIT	UNIT	3	0	Operation quantity unit	OP, quantity unit
OPER_QUAN_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used
OPER_QUANTITY	QUAN	13	3	Operation quantity	OP, target quantity (primary quantity)
PLANNED_SCRAP	QUAN	13	3	Scrap quantity of operation	OP, target scrap (primary quantity)
NOMINATOR	DEC	5	0	Numerator for the conversion from operation in planned quantity unit	Not used
DENOMINATOR	DEC	5	0	Denominator for the conversion from operation in planned quantity unit	Not used
UNDERDELIVERY_QUANTITY	QUAN	13	3	Quantity of the underdelivery tolerance in operation quantity unit	Conversion in percent: (UNDERDELIVERY_QUANTITY * 100) / OPER_QUANTITY

Field name	T	L	D	Description	Usage in HYDRA
CHECK_UNDERDELIVERY	CHAR	1	0	Check of the underdelivery tolerance in the order	" " No check "X" Error "W" Warning Note: You can only enter deviation reasons for overdeliveries/ underdeliveries via the CTWIN software. If you use DOS terminals, the reaction "W" is interpreted as an error ("X").
OVERDELIVERY_QUANTITY	QUAN	13	3	Quantity of the overdelivery tolerance in the operation quantity unit	Conversion in percent: (OVERDELIVERY_QUANTITY * 100) / OPER_QUANTITY
CHECK_OVERDELIVERY	CHAR	1	0	Check of the overdelivery tolerance in the order	" " No check "X" Error "W" Warning Note: You can only enter deviation reasons for overdeliveries/ underdeliveries via the CTWIN software. If you use DOS terminals, the reaction "W" is interpreted as an error ("X").
MESSAGE_TYPE	CHAR	1	0	Message type for checking the operation sequence	Not used
USERFIELD_CH20_1	CHAR	20	0	User field for 20 characters (SAP USR00)	Usage in HYDRA see below *2)
USERFIELD_CH20_2	CHAR	20	0	User field for 20 characters (SAP USR01)	Usage in HYDRA see below *2)
USERFIELD_UNIT	UNIT	3	0	User field: Unit of quantity field (SAP USE04)	Not used
USERFIELD_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used
USERFIELD_QUAN	QUAN	13	3	User field for the quantity (length 10,3) (SAP USR04)	Usage in HYDRA see below
ACTIVITY_UNIT_1	UNIT	3	0	Activity 1: Quantity unit of the activity to be uploaded.	Not used
ACTIVITY_UNIT_1_ISO	CHAR	3	0	Activity 1: ISO code of the quantity unit	Not used
ACTIVITY_QUANTITY_1	QUAN	13	3	Activity 1: Total activity quantity to be uploaded	Not used
ACTIVITY_UNIT_2	UNIT	3	0	Activity 2: Quantity unit of the activity to be uploaded.	Not used
ACTIVITY_UNIT_2_ISO	CHAR	3	0	Activity 2: ISO code of the quantity unit	Not used

Field name	T	L	D	Description	Usage in HYDRA
ACTIVITY_QUANTITY_2	QUAN	13	3	Activity 2: Total activity quantity to be uploaded	Not used
ACTIVITY_UNIT_3	UNIT	3	0	Activity 3: Quantity unit of the activity to be uploaded.	Not used
ACTIVITY_UNIT_3_ISO	CHAR	3	0	Activity 3: ISO code of the quantity unit	Not used
ACTIVITY_QUANTITY_3	QUAN	13	3	Activity 3: Total activity quantity to be uploaded	Not used
ACTIVITY_UNIT_4	UNIT	3	0	Activity 4: Quantity unit of the activity to be uploaded.	Not used
ACTIVITY_UNIT_4_ISO	CHAR	3	0	Activity 4: ISO code of the quantity unit	Not used
ACTIVITY_QUANTITY_4	QUAN	13	3	Activity 4: Total activity quantity to be uploaded	Not used
ACTIVITY_UNIT_5	UNIT	3	0	Activity 5: Quantity unit of the activity to be uploaded.	Not used
ACTIVITY_UNIT_5_ISO	CHAR	3	0	Activity 5: ISO code of the quantity unit	Not used
ACTIVITY_QUANTITY_5	QUAN	13	3	Activity 5: Total activity quantity to be uploaded	Not used
ACTIVITY_UNIT_6	UNIT	3	0	Activity 6: Quantity unit of the activity to be uploaded.	Not used
ACTIVITY_UNIT_6_ISO	CHAR	3	0	Activity 6: ISO code of the quantity unit	Not used
ACTIVITY_QUANTITY_6	QUAN	13	3	Activity 6: Total activity quantity to be uploaded	Not used
CO_BUSPROC_1	CHAR	12	0	Business process: Identification	Not used
BUSPROC_UNIT_1	UNIT	3	0	Business proc.: quantity unit of the quantity to be uploaded.	Not used
BUSPROC_UNIT_1_ISO	CHAR	3	0	Business process: ISO code of the quantity unit	Not used
BUSPROC_QUANTITY_1	QUAN	13	3	Business process: quantity to be uploaded	Not used
WORK_CNTR	CHAR	8	0	Workplace	OP, machine (see note)
PLANT	CHAR	4	0	Plant	Used in confirmations/uploads

Field name	T	L	D	Description	Usage in HYDRA
EARL_SCHED_START_DATE	DATS	8	0	Earliest scheduled start time: Execution (date)	OP Earliest start date OP Planned start (date)
EARL_SCHED_START_TIME	TIMS	6	0	Earliest scheduled start time: Execution (time)	OP Earliest start time OP Planned start (time)
LATE_SCHED_FIN_DATE	DATS	8	0	Latest scheduled end date: Execution (date)	OP Latest end date OP Planned end (date)
LATE_SCHED_FIN_TIME	TIMS	6	0	Latest scheduled end time: Execution (time)	OP Latest end time OP Planned end (time)
SETUP_TIME_UNIT	UNIT	3	0	Unit of setup time	HYDRA converts into internal format
SETUP_TIME_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used
SETUP_TIME	QUAN	7	1	Setup duration	OP Setup time (target BMK_7) *2)
PROCESS_TIME_UNIT	UNIT	3	0	Unit of processing time	HYDRA converts into internal format. The following units are supported: STD / H / HUR / HR
PROCESS_TIME_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used
PROCESS_TIME	QUAN	7	1	Duration of processing	OP Processing time (target BMK_11) *2)
TEARDOWN_TIME_UNIT	UNIT	3	0	Unit of tear down time/retooling time	Not used
TEARDOWN_TIME_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used
TEARDOWN_TIME	QUAN	7	1	Teardown duration/retooling duration	OP, target teardown time/retooling time
STATUS	CHAR	5	0	Individual status of an object	Not used
INDICATOR_TT_OR_TE	CHAR	1	0	Time ticket or time event	Not used
INDICATOR_FIN_OR_PART	CHAR	1	0	Partial finish or end status	OPs with end status ("E") will not be transferred
CONFIRMED_YIELD	QUAN	13	3	Total yield uploaded/confirmed in operation quantity unit	Not used
CONFIRMED_SCRAP	QUAN	13	3	Total scrap uploaded/confirmed in operation quantity unit	Not used
CONFIRMED_REWORK	QUAN	13	3	Total rework uploaded/confirmed in operation quantity unit	Not used
CONFIRMED_ACTIVITY_1	QUAN	13	3	Activity 1: Activity quantity already uploaded	Not used
NO_REMN_ACTIVITY_1	CHAR	1	0	Activity 1: Indicator: No remaining activity expected	Not used

Field name	T	L	D	Description	Usage in HYDRA
CONFIRMED_ACTIVITY_2	QUAN	13	3	Activity 2: Activity quantity already uploaded	Not used
NO_REMN_ACTIVITY_2	CHAR	1	0	Activity 2: Indicator: No remaining activity expected	Not used
CONFIRMED_ACTIVITY_3	QUAN	13	3	Activity 3: Activity quantity already uploaded	Not used
NO_REMN_ACTIVITY_3	CHAR	1	0	Activity 3: Indicator: No remaining activity expected	Not used
CONFIRMED_ACTIVITY_4	QUAN	13	3	Activity 4: Activity quantity already uploaded	Not used
NO_REMN_ACTIVITY_4	CHAR	1	0	Activity 4: Indicator: No remaining activity expected	Not used
CONFIRMED_ACTIVITY_5	QUAN	13	3	Activity 5: Activity quantity already uploaded	Not used
NO_REMN_ACTIVITY_5	CHAR	1	0	Activity 5: Indicator: No remaining activity expected	Not used
CONFIRMED_ACTIVITY_6	QUAN	13	3	Activity 6: Activity quantity already uploaded	Not used
NO_REMN_ACTIVITY_6	CHAR	1	0	Activity 6: Indicator: No remaining activity expected	Not used
CONFIRMED_BUS_PROC_1	QUAN	13	3	Quantity uploaded for the business process	Not used
NO_REMN_BUS_PROC_1	CHAR	1	0	No remaining quantity for the business process expected	Not used
CONFIRMED_WORK_CNTR	CHAR	8	0	Actual workplace	Not used
CONFIRMED_PLANT	CHAR	4	0	Actual plant	Not used

AK.xxx Field xxx in HYDRA order header

AG.xxx Field xxx in HYDRA operation

Structure E2BP_PP_PDC_OPERA1

The structure described in the following, controls the deletion process for already transferred production orders and/or their operations in the subsystem.

Field name	T	L	Description	Usage in HYDRA
SOURCE_SYS	CHAR	10	Logical system	Not used

Field name	T	L	Description	Usage in HYDRA
CONF_NO	NUMC	10	Confirmation/upload number of the operation	Confirmation/upload number
ORDERID	CHAR	12	Order	According to configuration (*1)
SEQUENCE	CHAR	6	Sequence	According to configuration (*1)
OPERATION	CHAR	4	Operation	According to configuration (*1)
SUB_OPER	CHAR	4	Suboperation	According to configuration (*1)
SUBSYSTEM_GROUPING	CHAR	3	Grouping of subsystem connection	Can be restricted to a subsystem group.

Notes on the operation structure

HYDRA order number (*1)

The HYDRA order number consists of a configurable part of the SAP key fields ORDERID, SEQUENCE, OPERATION, SUB_OPER. The total length must not exceed 25 digits. If DOS terminals are used, the total length must not exceed 16 digits.
When the CHAR fields are transferred, the system converts blank characters to "0", since the barcode is also assigned "0" instead of blank characters.

Overview - usage of user fields

By default, the user-specific fields USERFIELD_CH20_1, USERFIELD_CH20_2 and USERFIELD_QUAN are processed in HYDRA as follows:

User field 01 (USERFIELD_CH20_1):

Places 1…10:
--> BDE cycle time in seconds per 1000 machine strokes in the format NUMC 10

Places 11…20:
--> not used in HYDRA

User field 02 (USERFIELD_CH20_2):

Places 1…10:
--> Premium default target te in seconds per 1000 pieces in the format NUMC 10

Places 11…20:
--> Premium default target tr in seconds in the format NUMC 10

User field 04 (USERFIELD_QUAN):

→ HYDRA partitioning (parts per cycle)

HYDRA partitioning

If the user field USERFIELD_QUAN has a value this value will be entered as the partitioning in the operation.

If this is not the case, the system implicitly assumes a partitioning of 1 and enters this value in the operation.

BDE cycle time

The system attempts to read the cycle time from user field 01 - places 1...10. If this is possible, the system enters this value as the target cycle time in the HYDRA operation.

If the system cannot identify the target cycle time as described above, then the system calculates it from the specified processing time (PROCESS_TIME) and the partitioning (TEILIGKEIT) values that might be specified.

Calculation basis (formulas):

Processing time OP = Target quantity OP * (target cycle time machine/ partitioning)

- Target cycle time machine (= BDE cycle time/ machine stroke)
- Target cycle time/ unit = target cycle time machine/ partitioning

=> Target cycle time machine= (processing time OP * Partitioning)/ target quantity OP

Standard times

In HYDRA the content of the field "processing time" (PROCESS_TIME) specifies the standard time for machine assignment (target for RPA 11).

In HYDRA the content of the field "Setup time" (SETUP_TIME) specifies the standard time for the machine setup (target for RPA 7).

Note on the workplace field WORK_CNTR:

As of version 6.5 HYDRA supports alphanumeric machine/workplace numbers. In this case, you can only use the Windows terminals CT 76x and CT 8xx for data collection.

Notes on all other alphanumeric fields:

HYDRA does not support specific special characters for all alphanumeric fields. These characters are: "%", "\", "/", "|" as you cannot enter these characters using shop floor terminals; the terminals and the MOC do not support these characters. Do not use the characters ";", " “ ", and " ’ " since they are often interpreted as comment characters or separators and will thus lead to unwanted effects.